KARTHIKEYAN K

Data Scientist / AI ML Engineer

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| Github/kkarthik3 | LinkedIn/kkarthick-k |Portfolio|

PROFESSIONAL SUMMARY

Machine Learning Engineer with 3.5+ years of professional experience, including 2.5 years in AI/ML and 1 year in Python data operations. Experienced in designing and deploying RAG-based chatbots, vector search engines, Multi Agent Orchestration and real-time computer vision systems using LangChain, LangGraph, Neo4j, FastAPI, and cloud platforms such as AWS and Kubernetes. Skilled in data preprocessing, model development, and scalable deployment, delivering production-ready AI solutions that drive efficiency and improve user experiences across US Healthcare, e-commerce, and Automotive domains.

EXPERIENCE

AI/ML Engineer [Value Health Solutions, Coimbatore] **Dec 2025 – Present**

- **AI-Driven SDLC & Intelligent Project Management:** Developed an AI-powered SDLC tracking and project management solution integrating Jira and Microsoft ecosystem services through enterprise connectors and Model Context Protocol (MCP)-based tool integrations. Enabled AI agents to interact with project management tools, retrieve contextual information, automate task-related workflows, and consolidate project status, dependencies, and delivery progress across enterprise systems. Designed tool-calling workflows to support AI-assisted project monitoring and development lifecycle management.
- **Multi-EHR AI Interoperability Platform - Medivo:** Contributed to the development in Medivo, a healthcare interoperability platform that integrates with multiple Electronic Health Record (EHR) systems using SMART on FHIR and HL7 FHIR standards. Enabled secure connectivity with healthcare systems to retrieve and consume over 200+ real-time clinical data through FHIR APIs, supporting interoperability across disparate EHR platforms and facilitating access to patient information for downstream healthcare workflows.
- **Graph RAG Pipeline Development:** Introduced and led adoption of Graph Retrieval-Augmented Generation (Graph RAG) for enterprise RAG applications, designing a scalable knowledge graph–based retrieval pipeline with custom components for AWS Bedrock, Textract, and domain-specific data loaders to enable context-aware, multi-hop reasoning.
- **AI-Powered Revenue Cycle Management (RCM) Application:** Developed an AI-powered Revenue Cycle Management (RCM) dashboard that automates patient intake processing using OCR and AI agents, extracting and validating clinical and billing information from patient intake forms. Built intelligent workflows to calculate medical billing and cost estimation using CPT, ICD, and diagnosis codes, improving billing accuracy and operational efficiency. Designed AI agents to identify and map doctor information directly from patient intake forms and clinical records. Integrated and collaborated with 10+ US healthcare payers to streamline claims processing, eligibility verification, and billing workflows across healthcare operations.

Machine Learning Engineer[Kadit Innovations, Tenkasi, India] **Mar 2025 – Nov 2025**

- **Protein Vector Search :** Designed and implemented an AI-powered protein search engine using Neo4j vector databases to perform similarity-based retrieval, integrating metadata from multiple protein databases to map sequences and identify relevant matches, including detailed annotation of post-translational modifications (PTMs).
- **CCTV Attendance Monitoring System**: Built a real-time attendance and security monitoring solution using RTSP camera feeds with on-device motion detection. Integrated WebSocket triggers to capture events, applied YOLO for person detection and InsightFace facial embeddings for identity recognition, and logged verified entries into PostgreSQL for attendance tracking and auditing.
- Deployed the applications on AWS, leveraging services such as EC2, S3, EKS and Lambda to ensure scalable performance and reliability for processing large biological datasets in a cloud-native environment.

Associate Machine Learning Engineer [ISPG Technologies Pvt.Ltd, Kochin, Kerala] **Dec 2024 – Mar 2025**

(Intern: 7 Months → Associate ML Engineer: 3 Months)

- **RAG-Based Product User-Manual Chatbot (Development)**: Created a RAG-based product manual chatbot API by crawling and scraping over 5000 websites using Selenium, stored embeddings in Mongo-VectorSearch, and structured the workflow using LangGraph with tools and agents for particular sections.
- **AI Chatbot Workflow with LangGraph Agents & AWS (Pre-Production):** Developing a product with a structured workflow for an AI chatbot using LangGraph tools and agents, enabling efficient, task-specific interactions by automating responses and decision-making processes, improving user experience and achieving accuracy levels

of over 70% compared to manual flow approaches. The solution features a centralized chatbot with an admin dashboard for controlling and customizing functionalities.

- **RAG Technology Integration (Development):** Developed and integrated Database Retrieval-Augmented Generation (RAG) into the CRM database, enabling seamless translation of user NLP queries into Cypher queries (Graph DB) and converting Cypher responses into human-readable formats. Implemented restricted delete, create, and update operations. This integration enhanced client interaction and operational efficiency by 40%.
- **AI-Based E-Commerce Query System (Production):** Designed and developed two AI-based e-commerce query systems and an admin dashboard for cost and query analysis using MongoDB VectorSearch for automobile spare parts and jewelry retailers. Implemented AI-driven search systems that enhanced customer experience with fast, accurate queries and natural language processing by 200ms. Integrated advanced search and AI recommendations, improving user experience by 50% and driving sales.

Data & Operations Associate [MK Labels, Chennai**]** **May 2021 – Jul 2022**

- Collected, organized, and maintained large datasets of financial transactions (credit, debit, accounts payable/receivable) using Excel and Pandas, ensuring data accuracy and integrity.
- Managed inventory Data of 500,000+ stock units/Day using MySQL, ensuring 99% on-time delivery and accurate stock records.
- Streamlined workflows between clients and production teams using MySQL, Cron jobs, and automated email alerts, increasing operational efficiency by 20%.

EDUCATION

Bharathiar University, Coimbatore **Aug 2022 - Apr 2024**

Masters in Data Analytics, Cumulative CGPA: 7.91/10

Ramakrishna Mission Vivekananda College, Chennai **Jun 2018 - Jun 2021**

Bachelors in Physics, Cumulative CGPA: 8.02/10

SKILLS

Libraries	Python (Numpy, Pandas, Matplotlib, Seaborn, Scikit-Learn, Tensorflow, Keras, NLTK, Streamlit, Fast-API, Flask, Langchain, Langgraph, Neo4j, Deepeval
Languages	Python ,SQL, Cypher (Neo4J)
Machine Learning	Data Modeling, Clustering & Classification, Quantitative Analysis, Regression, Transformers, Predictive Modeling, Model Validation, Model deployment, CNN & RNN, LSTM, BERT.
Artificial Intelligence	AI model Traning, FineTuning, AI Evaluation, HybridRetrieval Augement Genreration (RAG), VLLM, AI Model serving, Multi Agent Orchestration, Harness Engineering
Tools	Jupyter Notebook, Arduino IDE, MS Excel, Tableau, VS code, Postman, n8n.
Technologies	Langchain, LLM, Large Language Models, IoT, Onnx, Docker, Kubernetes, Langfuse
Pipleines	Kedro Pipelines, Github actions Pipelines, Kubeflow Pipelines
Cloud	***Amazon Web Service (***AWS EC2, AWS IAM AWS Bedrock, AWS RDS, AWS APIGateway, AWS Lambda, AWS S3, AWS ECR), GCP (**Google Cloud Platform**)

COURSES AND CERTIFICATIONS

- Neo4j Certified Professional, **Neo4j**
- AWS Certified AI Practitioner AIF-C01, **Udemy**
- AWS Certified Cloud Practitioner CLF-C02, **Udemy**
- Cypher Fundamentals, **Neo4J**
- Database Management System, **NPTEL**